FinBuddy: Explainable Financial Scam Detection and Education Using BERT, BiLSTM, SHAP, and RAG

Group 6: Nikhilesh Dhavale, Aditya Karande, Simintaj Salehpour, Nithin Ravindra Reddy

1. Problem Selection and Justification

Financial scams and phishing messages pose significant risks to consumers. This project proposes FinBuddy, an explainable NLP-based assistant that detects suspicious messages, explains model predictions, and provides reliable financial safety guidance. The project will compare traditional machine learning, recurrent neural networks, and transformer-based models in terms of classification performance, interpretability, and robustness. FinBuddy will integrate classification, explainability, and Retrieval-Augmented Generation (RAG) into a unified workflow.

2. Dataset

The primary dataset will be the SMS Phishing Dataset (Mendeley), containing 5,971 messages labeled as ham, spam, or smishing. The SmishTank dataset, containing 1,090 real-world smishing messages, will support external validation. Financial-related messages will be identified, and duplicate records will be removed to prevent data leakage. The RAG knowledge base will use trusted resources from CFPB, FTC, and SEC Investor.gov.

3. NLP Methods and Models

Three classification approaches will be compared: TF-IDF with Logistic Regression (classical baseline), BiLSTM (recurrent neural network), and fine-tuned BERT (pretrained transformer). SHAP will explain influential words and phrases contributing to model predictions. The classification results and explanations will guide the RAG component, which will use sentence-transformer embeddings, FAISS, and an LLM to retrieve relevant financial safety information and generate source-grounded responses.

4. Framework and NLP Tasks

The project will use Python, PyTorch, Hugging Face Transformers, scikit-learn, SHAP, Sentence Transformers, FAISS, and Streamlit. The main NLP tasks include text classification, explainable prediction, robustness analysis, semantic retrieval, and retrieval-augmented question answering.

5. Performance Evaluation

Classification models will be evaluated using Precision, Recall, Macro-F1, PR-AUC, and confusion matrices. Robustness will be assessed using modified and paraphrased messages. External validation will evaluate the ability to identify independent examples. RAG performance will be assessed through retrieval relevance, answer correctness, and faithfulness to retrieved sources.

Preliminary Schedule: Week 1: Data preparation and RAG source collection; Weeks 2–3: Model development and comparison; Week 4: SHAP, robustness testing, and RAG implementation; Week 5: System integration, evaluation, final report, and presentation.